Contribution Journal

Source File:

MT_Contribution_Journal_Alex_DeBusk.pdf

Individual Contribution

For this midterm project, my contribution focused on understanding the project requirements, planning the data preprocessing workflow, and preparing the project documentation. I reviewed the instructor's guidelines and the proposed approach for cleaning and preparing the CarDekho dataset for machine learning.

I also worked on the project proposal, outlining the dataset's purpose, the preprocessing techniques to be used, and the machine learning models planned for the next stage of the project. This included understanding the importance of handling missing values, encoding categorical variables, scaling numerical features, and splitting the dataset into training and testing sets to prevent data leakage.

Additionally, I reviewed the existing preprocessing work to better understand how the dataset was prepared for machine learning and what would be required for the final submission. This helped me become more familiar with the overall workflow and the role of data preprocessing in building regression models to predict used car prices.

Through this project, I gained a better understanding of exploratory data analysis, data preparation, and the importance of organizing a dataset before applying machine learning techniques. My contribution was primarily focused on project planning, documentation, and reviewing the preprocessing approach.

Contribution Journal

Source File:

MT_Contribution_Journal_Shareefah_Adeola_ITAI1371.pdf

Contribution Journal

Student: Shareefah Adeola

Team: ML_1371_14274 4_ITAI1371

Date: 10/03/2026

My Contribution to the Used-Car Machine Learning Project

As a member of the project team, I contributed to the preparation, organization, and preprocessing of the used-car dataset.

My contributions included:

1. Dataset organization:

   I participated in organizing the original dataset and helped introduce vehicle type as a grouping variable. This made the vehicle information more manageable for our analysis, particularly because the original dataset contained many vehicle makes/categories.

2. Data cleaning:

   I participated in identifying and removing duplicate records and identifying missing values in the dataset.

3. Missing-value treatment:

   I worked with the team to replace missing numerical values using the median of the appropriate training-data variables.

4. Outlier analysis and treatment:

   I participated in the IQR analysis used to identify extreme values. For km_driven, the upper IQR boundary was calculated and the extreme values were capped rather than removing those observations.

5. Exploratory Data Analysis:

   I participated in examining the distributions and relationships among variables such as selling price, year, kilometer driven, engine, maximum power, mileage, fuel type, transmission, and ownership.

6. Feature engineering:

   I contributed to creating the vehicle_age feature from the vehicle year, using 2020 as the reference year.

7. Categorical data preprocessing:

   I worked with the team on preparing categorical variables for machine learning using one-hot encoding.

8. Numerical feature scaling:

   I participated in standardizing numerical features using StandardScaler so that the numerical predictors were placed on a comparable scale.

9. Training and testing data preparation:

   I learned and participated in maintaining separate training and testing datasets and applying preprocessing consistently using the training-data parameters.

10. Documentation and project submission:

    I contributed to documenting the preprocessing process, preparing the reflection, and organizing the files required for the project submission.

Through these contributions, I gained practical experience in preparing a real-world dataset for machine-learning analysis. The project also strengthened my understanding of Python, Jupyter/Google Colab, data cleaning, exploratory analysis, feature engineering, encoding, and scaling.

Contribution Journal

Source File:

MT_Contribution_Journal_Yilin_Leng_ITAI1371.pdf

Individual contribution

For this midterm project, I was mainly responsible for data cleaning, exploratory data analysis (EDA), and preprocessing the used car dataset for machine learning. I independently worked through the main data processing workflow, from loading the raw dataset and checking the data to preparing the final processed dataset. I first loaded the raw data into a Jupyter Notebook using Python and checked its structure, data types, missing values, and the distributions of the main numerical variables. Following the instructor's requirements, I completed all data processing through Python code instead of manually changing the original dataset. I split the dataset into 70% training data and 30% testing data, and I performed the EDA and preprocessing only on the training data while keeping the testing data separate to avoid data leakage.

During the analysis, I found that the used car price variable had a strong right skew, so I applied a logarithmic transformation to improve its distribution. I also standardized the continuous numerical features and encoded the categorical variables so that the data could be used more effectively by machine learning models. In addition, I used Matplotlib to visualize the data distributions before and after preprocessing and printed sample data at different stages to check whether the processing results were correct. Through this project, I improved my skills in using

Python, Pandas, and Matplotlib for data analysis and preprocessing, while also gaining a better understanding of why training and testing data need to be kept separate and how data leakage can affect machine learning results. Overall, my work helped prepare the dataset for the next stage of the project, where multiple machine learning models will be trained and evaluated.